

Tianlong Wang

PhD candidate · Zhejiang University

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Seeking postdoctoral opportunities · Expected PhD completion: June 2027 · Available from July 2027

Research profile

I develop physics-informed and explainable machine-learning methods for debris-flow, landslide and hydrological prediction. My research combines catchment storage and hydrological memory with residual learning, ensemble optimization and uncertainty analysis, with particular attention to small catchments and sparse observations. I seek to extend these methods to related problems in hydrology, geotechnics and engineering monitoring through interdisciplinary collaboration.

Research outputs: 19 published papers (18 journal articles and 1 conference paper); 5 manuscripts under review; 3 investigator-led research awards totalling RMB 126,000; 6 open-source research software projects.

Education

Zhejiang University · September 2022 – June 2027 (expected)

PhD · Ocean Technology and Engineering · Ocean College

Supervisor: Hongyue Sun

Nanyang Technological University · September 2024 – September 2026

Joint PhD Student · School of Civil and Environmental Engineering

Supervisor: Chu Jian

Research competencies

- **Physics-informed prediction:** Water-balance reasoning, hydrological memory and physical consistency constraints for runoff, water-stage and landslide forecasts.
- **Explainable AI and algorithm development:** Ensemble learning, metaheuristic optimization, SHAP analysis and pretrained tabular models for geohazard assessment and sparse-data prediction.
- **Evaluation and uncertainty:** Chronological and spatial evaluation, robust state estimation and sequential prediction intervals.

Selected manuscripts under review

1. **Physics-guided residual stacking for short-lead runoff and stage forecasting in the Fengshuwan catchment, China**
Tianlong Wang, Jian Chu, Hao Yang, Hongyue Sun*
[Journal of Hydrology: Regional Studies](#) · Under review following revision
2. **Basin Water Storage Degree for Risk-Informed Debris-Flow Warning Support: A Forward-Looking Proof of Concept in a Small Mountain Catchment**
Tianlong Wang, Jian Chu, Hao Yang, Hongyue Sun*
[International Journal of Disaster Risk Reduction](#) · Under review following revision
3. **Hydrological readiness forecasting for debris flow early warning in a mountain catchment with sparse event records**
Tianlong Wang, Jian Chu, Jingkun Su, Runting Fan, Hao Yang, Hongyue Sun*
[Bulletin of Engineering Geology and the Environment](#) · Under review
4. **Robust state estimation and hydrology-informed residual learning for landslide velocity forecasting with prequential prediction intervals**
Tianlong Wang, Jian Chu, Hao Yang, Hongyue Sun*
[Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards](#) · Under review
5. **Sediment-source information improves small-sample prediction of maximum recorded debris-flow volume with a pretrained tabular model**
Tianlong Wang, Jian Chu, Yanhui Li, Hao Yang, Hongyue Sun*
[Stochastic Environmental Research and Risk Assessment](#) · Under review following revision

Selected published articles

1. **DOME: A Dynamic Optimization Meta-Ensemble Framework for Debris-Flow Susceptibility Assessment with SHAP-Based Interpretation**
Tianlong Wang^{*}, H. Yang
Natural Hazards, 122, Article 545. · 2026 · DOI
2. **Intelligent Prediction of Post-Seismic Debris Flow Source Volumes: A Physics-Informed Generative Deep Learning Framework**
Tianlong Wang, H. Yang, H. Sun^{*}
Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards, 1–29. · 2026 · DOI
3. **An integrated IKOA-CNN-BiGRU-Attention framework with SHAP explainability for high-precision debris flow hazard prediction in the Nujiang river basin, China**
H. Yang, Tianlong Wang^{*}, N. I. Fomin, S. Xiao, L. Liu
PLOS ONE, 20(6), e0326587. · 2025 · DOI
4. **A novel method for predicting debris flow hazard: a multi-strategy fusion approach based on the light gradient boosting machine framework**
Tianlong Wang, Q. Ge, T. Ma, et al.
Stochastic Environmental Research and Risk Assessment, 39, 4867–4890. · 2025 · DOI
5. **Prediction and explanation of debris flow velocity based on multi-strategy fusion Stacking ensemble learning model**
Tianlong Wang, K. Zhang, Z. Liu, T. Ma, R. Luo, H. Chen, X. Wang, W. Ge, H. Sun^{*}
Journal of Hydrology, 638, 131347. · 2024 · DOI
6. **Study and verification on an improved comprehensive prediction model of landslide displacement**
Tianlong Wang, R. Luo, T. Ma, et al.
Bulletin of Engineering Geology and the Environment, 83, Article 90. · 2024 · DOI
7. **Indoor Model Test of Negative Pressure Drainage Method for Treatment of Engineering Mud**
Z. Chu, S. Yu, X. Li, X. Lu, X. Wang, R. Luo, Tianlong Wang^{*}
Journal of Changjiang River Scientific Research Institute, 41(9), 114–122, 129. (In Chinese) · 2024 · DOI
8. **Study on wavelet multi-scale analysis and prediction of landslide groundwater**
Tianlong Wang, D. Peng, X. Wang, B. Wu, R. Luo, Z. Chu, H. Sun^{*}
Journal of Hydroinformatics, 26(1), 237–254. · 2024 · DOI

Honors & awards

National Scholarship · 2025, 2021

China

Outstanding Graduate · 2022

Chinese Society for Rock Mechanics and Engineering · One of 100 recipients nationwide

Silver Award · 2025

China International College Students' Innovation Competition, Singapore Regional

Grand Prize · 2021

The 3rd National College Student Structural Design Information Technology Competition, China

Research funding & projects

Young Elite Scientists Sponsorship Program Doctoral Student Special Plan

China Association for Science and Technology · **PI** · January 2025 — December 2026

Lead research on prediction methods and model interpretation for small-catchment debris-flow studies.

Project ID: 156-O-170-0000603-2 Funding: RMB 40,000.

Seed Fund Cultivation Project of Ocean College

Zhejiang University · **PI** · June 2025 — June 2027

Develop a doctoral research programme connecting catchment hydrological states with physics-guided prediction.

Grant No. 2025BS004 Funding: RMB 50,000.

Outstanding Doctoral Dissertation Award Cultivation Grant

Zhejiang University · **PI** · 2025–2026

Doctoral research support for integrating hydrological-state analysis, prediction and interpretation.

Grant No. 825011A · RMB 36,000.

Key Program

National Natural Science Foundation of China · **Core team member** · January 2023 — December 2027

Contribute to the project's hydrology-informed geohazard prediction research; related work includes storage-state and short-lead forecasting studies.

Grant No. 42230702 Total project budget: RMB 2,730,000.

Research software

DOME · Repository ↗

Dynamic ensemble selection and optimization for regional debris-flow susceptibility, with SHAP interpretation.

TabPFNV2.5-DebrisFlow · Repository ↗

Small-sample debris-flow volume prediction using a pretrained tabular model and sediment-source information.

RSRL-HU · Repository ↗

One-day-ahead landslide velocity forecasts with hydrology-informed residual learning and sequential intervals.

Hydrological Readiness Forecasting · Repository ↗

Three-day probabilistic forecasts of warning-relevant hydrological states, evaluated chronologically.

BWSD · Repository ↗

Water-balance-based catchment storage indicator and daily hydrological-state classification.

HydroPIML-RS · Repository ↗

Physics-guided residual stacking for short-lead runoff and water-stage prediction.

Academic service

Selected journal reviewing: Journal of Hydrology; Acta Geotechnica; Bulletin of Engineering Geology and the Environment; Stochastic Environmental Research and Risk Assessment; Applied Intelligence; Array; International Journal of Interactive Multimedia and Artificial Intelligence.

Memberships: Chinese Society for Rock Mechanics and Engineering; Geological Society of China; China Civil Engineering Society.

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